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Defying the Speed of Light: How Predictive Temporal AI Architectures Sustain Social Presence and Relational Cohesion in Deep Space Communication

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ABSTRACT

Communication latency between Earth and deep-space crewed spacecraft introduces a fundamental breakdown in collaborative decision-making and interpersonal social presence. For a Mars mission at median opposition, one-way signal travel time reaches $m \approx 240$ seconds, rendering conventional synchronous computer-mediated dialogue operationally and socially impossible. This paper proposes and formally evaluates the *Predictive Temporal Bridge* (PTB)—a dual-AI architecture that transforms asynchronous, lag-driven telemetry exchanges into a functionally synchronous social and decision-support channel. The PTB deploys an *Earth AI* that projects received spacecraft state forward by $2m$ seconds using physics-based predictive models, and a *Cabin AI* that reconciles Earth's forward projection against real sensor truth upon the message's arrival. The result is a command packet that arrives at the spacecraft at the exact moment it was computed for, providing the Human Commander with the perceptual, psychological, and operational experience of near-real-time collaboration. Crucially, we ground our evaluation in computer-mediated communication (CMC) theories by introducing and analyzing four vital Human-Computer Interaction (HCI) metrics. Using a high-fidelity interactive simulation of a Mars orbital-insertion burn anomaly, we measure the *Social Presence Index* (SPI) to quantify the crew's perception of “being there” with Ground Control; the *Cognitive Load Index* (CLI) to evaluate operational stress during predictive desynchronization; the *Conversational Latency Illusion* (Lat) to measure the system's effectiveness in perceptually masking the true multi-minute signal delay; and the *Shared Situational Awareness* (SA) Sync Rate to validate the alignment between Earth's mental model and Mars' physical reality. Our results demonstrate that the PTB successfully sustains a seamless “co-pilot” dynamic, preventing psychological detachment, minimizing cognitive fatigue, and ensuring human autonomy via a robust reality-validation engine, thereby bridging the profound sense of isolation characterizing long-duration deep space exploration.

1 | Introduction

Every crewed deep-space mission must confront a communication constraint that is, unlike most engineering problems, non-negotiable: the finite speed of light. At Mars opposition, electromagnetic signals require between 182 and 1,342 seconds one-way, with a median value near 240 seconds [1]. This delay, denoted m , means that any question posed by a Mars crew member will not

reach Earth for m seconds, and any Earth response will take another m seconds to return—a total round-trip latency of $2m \approx 8$ minutes at median separation, growing up to 45 minutes at maximum distance.

The consequences cascade far beyond simple system performance; they disrupt the core social fabric of human collaboration. In human-to-human systems under extreme stress, relational maintenance and mutual responsiveness are essential [2]. When

latency breaks the synchronous loop, conversational flows degenerate into isolated monologue transmissions. The psychological impact on the crew is a severe sense of temporal displacement and isolation from the human species [3]. [4] demonstrated in analog Mars missions that communication delays up to 22 minutes measurably elevate Mission Controller workload, anxiety, and stress, with error rates doubling in off-nominal scenarios. Conversely, [5] showed that integrating human-in-the-loop AI can alleviate performance degradation, but did not address the relational and social breakdowns that occur under delayed conditions.

Traditional space engineering has responded to this challenge by advocating for unilateral crew autonomy, pushing onboard systems to make the crew entirely independent of ground support. While technically logical, this "autonomy bubble" induces profound relational decay: crews begin to perceive Earth Mission Control as progressively out of touch, leading to friction, psychological reactance, and mission-threatening isolation [3]. Redesigning the *computer-mediated interface* to create an adaptive temporal architecture is an urgent, human-centered alternative. By using dual-AI predictive modeling to construct a virtual "present," we can bypass the cognitive and relational stutter of multi-minute lag.

This paper introduces the **Predictive Temporal Bridge (PTB)**, an AI-mediated communication (AI-MC) architecture designed to overcome this temporal distance. Following the foundational agenda of [6], we define AI-MC as communication where an AI agent acts on behalf of or transforms the communicative exchanges between human actors. The PTB represents a radical expansion of AI-MC: rather than modifying verbal content or generating automated text suggestions, the AI synchronizes the temporal frame of reference of the human interlocutors. The human commander on Mars receives ground commands that feel instantaneous and perfectly synchronized with their immediate physical state, restoring a feeling of synchronous presence.

2 | Theoretical Framework

2.1 | Social Presence and Interpersonal Latency in CMC

Social Presence Theory [7] posits that communication media vary in their capacity to transmit the psychological salience of the other person in the interaction. A critical determinant of social presence is response latency. In interpersonal interactions, a delayed response is often interpreted as non-verbal cue of disinterest, distance, or cognitive friction [8]. In deep-space environments, this latency is not a choice but a physical constraint, yet the human mind continues to experience it as a relational barrier. The absence of immediate turn-taking disrupts the "conversational grounding" process—the continuous exchange of mutual understanding and feedback that binds collaborative teams [2]. By eliminating the round-trip latency loop through algorithmic forward-projection, the PTB seeks to maintain high levels of social presence, allowing the Mars crew to feel psychologically tethered to the ground team.

2.2 | Hyperpersonal Mediation and the Illusion of Co-Presence

The Hyperpersonal Model of CMC [8] argues that computer-mediated environments can sometimes exceed face-to-face interactions in relational warmth and efficiency, due to selective self-presentation, partner idealization, and asynchronous editing. The PTB leverages this hyperpersonal dynamic by transforming an asynchronous, high-stress environment into a curated "virtual present." Because the Earth AI is constantly projecting the ship's state forward, the ground operators are enabled to respond to the commander's needs before the commander has even fully articulated them. This creates a hyperpersonal "co-pilot" effect: ground controllers appear uniquely prescient and supportive, enhancing mutual trust and collective efficacy despite a 400-million-kilometer separation.

2.3 | Predictive Displays and Time-Delay Teleoperation

The concept of leveraging prediction to mask latency has robust origins in robotics and space teleoperation. To mitigate the severe mechanical instability caused by transmission delays, engineers have long employed *predictive displays* and algorithmic forward-projection models, such as the Smith Predictor [9]. A prominent historical example is the "phantom robot" overlay developed at NASA JPL, which projects a simulated, real-time graphic of a robotic arm over a delayed video feed, allowing the human operator to visually confirm their intended motion without waiting for the physical round-trip signal [10].

While these classical predictive displays successfully close the mechanical control loop, they treat the human operator in isolation, focusing exclusively on human-machine mechanical teleoperation. The Predictive Temporal Bridge (PTB) elevates this paradigm from the realm of mechanical control into the domain of computer-mediated social interaction and collaborative decision-making. Rather than merely predicting the spatial position of a robotic arm to aid a single operator, the PTB leverages dual-AI predictive modeling to synchronize the entire temporal frame of reference between *two* human operational teams (Earth and Mars). By pairing physical telemetry projection with a human-in-the-loop reality reconciliation engine, the PTB ensures that deep space communication retains the psychological immediacy and social presence of real-time dialogue, while explicitly managing the cognitive load and ethical autonomy of the crew.

2.4 | Algorithmic Mediation, Agency, and Ethical Responsibility

Deploying AI to intercept and predict human communicative intent introduces major ethical questions concerning agency and automation bias [6]. When the Earth AI projects the ship's state and suggests a correction command, it is operating in a highly sensitive cognitive corridor. If the cabin crew blindly accepts these recommendations, true human agency is eroded—a phenomenon known as automation bias. Furthermore, if a predictive command is executed and fails, the moral accountability is diffused: does responsibility lie with the Earth flight director who approved the prediction, the Cabin AI that reconciled it, or the

commander who clicked "Execute"? The PTB explicitly addresses this ethical dilemma by implementing a dual-gate architecture. The Cabin AI is designed as a *reality arbiter* that compares ground predictions against local sensor reality, highlighting the exact mathematical variance. By making the AI's predictive assumptions completely transparent and subject to local human veto, the PTB preserves crew autonomy and ensures that ultimate moral and operational responsibility remains anchored in human agency.

2.5 | Shared Situational Awareness in Distributed Space Teams

[11] defines situational awareness (SA) as the perception, comprehension, and projection of environmental states. In high-risk collaborative environments, teams must establish a *shared SA*—a compatible mental model of the system's current and future states [12]. With raw light-time delay, shared SA is physically shattered: Earth Mission Control is stuck viewing the past ($t - m$), while the Mars crew resides in the present (t). The PTB resolves this temporal asymmetry by utilizing the Earth AI to mathematically project Earth's perception forward to the crew's immediate future ($t + m$), re-establishing a shared temporal anchor and a synchronized mental model.

3 | The Predictive Temporal Bridge Architecture

The PTB consists of three distinct components: the Mars Broadcast Module, the Earth Predictive Engine, and the Cabin Reconciliation Engine. The overall architecture is modeled as a dynamic time-synchronized feedback loop.

3.1 | Mars Broadcast Module

At spacecraft time t , the Cabin AI continuously packages the physical state $\mathcal{J}(t)$ along with the crew's operational intent vector $\vec{I}(t)$ and system noise envelopes $\Sigma(t)$ into a broadcast packet $\mathcal{B}(t)$:

$$\mathcal{B}(t) = \{\mathcal{J}(t), \vec{I}(t), \Sigma(t)\}. \quad (1)$$

The intent vector $\vec{I}(t)$ is a high-fidelity representation of the crew's current actions, planned thruster sequences, and behavioral metrics, providing the Earth AI with the contextual constraints necessary for accurate forward projection.

3.2 | Earth Predictive Engine

Upon receiving $\mathcal{B}(t)$ at Earth clock $t + m$, the Earth AI immediately projects the spacecraft state forward by $2m$ seconds to estimate its state at the exact moment the ground command will arrive back at Mars:

$$\hat{\mathcal{J}}(t + 2m) = \mathcal{J}(t) + \int_t^{t+2m} f(\mathcal{J}(\tau), \vec{I}(t)) d\tau + \epsilon, \quad (2)$$

where $f(\cdot)$ represents the physics-based system dynamic equations (including orbital mechanics, thermal dissipation, and thruster erosion profiles), and ϵ represents the uncertainty

bounds. Ground controllers review this predicted state on a real-time predictive dashboard, enabling them to issue correction commands $\mathcal{C}$ for anomalies that have not yet physically manifested on the ship.

3.3 | Cabin Reconciliation Engine

When the command $\mathcal{C}$ arrives at Mars at time $t + 2m$, the Cabin AI compares the actual physical telemetry $\mathcal{J}(t + 2m)$ with Earth's prediction $\hat{\mathcal{J}}(t + 2m)$:

$$\Delta_k = T_k(t + 2m) - \hat{T}_k(t + 2m), \quad k \in \text{telemetry channels}. \quad (3)$$

If the deviation Δ_k is within the safe margin (δ_{safe}), the Cabin AI validates the command as RECONCILED and presents it to the commander with a green recommendation badge. If the deviation is excessive (e.g., due to an unmodeled local event like a micrometeoroid strike), the Cabin AI flags the state as DELTA-HIGH, blocks auto-execution, and recommends manual override.

3.4 | Hardware Realization: The Predictive Temporal Processing Unit (PTPU)

To ensure deterministic execution and satisfy rigorous Aerospace Technology Journal standards for deep-space avionics, the PTB is physically implemented via a specialized Predictive Temporal Processing Unit (PTPU). The PTPU is a low-power Application-Specific Integrated Circuit (ASIC) that acts as an AI co-processor interfacing directly with the spacecraft's Primary Flight Computer (PFC).

As illustrated in Figure 3, the PTPU internal architecture couples a deterministic RISC-V lockstep CPU with a dedicated Neural Matrix Accelerator (NMA) for rapid forward-projection calculations. All PTB logic and kinematic models are immutably flashed onto non-volatile firmware ROM, shielding the reality arbitration algorithms from single-event upsets (SEUs) caused by cosmic radiation. State buffers are held in radiation-hardened SRAM.

At the network level (Figure 4), the PTPU communicates with spacecraft sensors, the PFC, and the Commander's Human-Machine Interface (HMI) dashboard via a universal SpaceWire or MIL-STD-1553 data bus. This isolation ensures that the predictive AI models never directly command thruster actuators without explicit authorization routed through the Commander HMI's reality overlay.

4 | Architecture of the Design

The dashboard splits the workspace into two primary columns, establishing a clean visual separation of the communicative partners. Figure 5 illustrates the high-fidelity split-screen view. The interface is divided into the following functional panels:

Earth Command Center (Left Panel): Represents the view from Ground Control.

- *Telemetry Tab:* Shows the delayed, stale telemetry exactly as it is received from Mars ($t - m$).

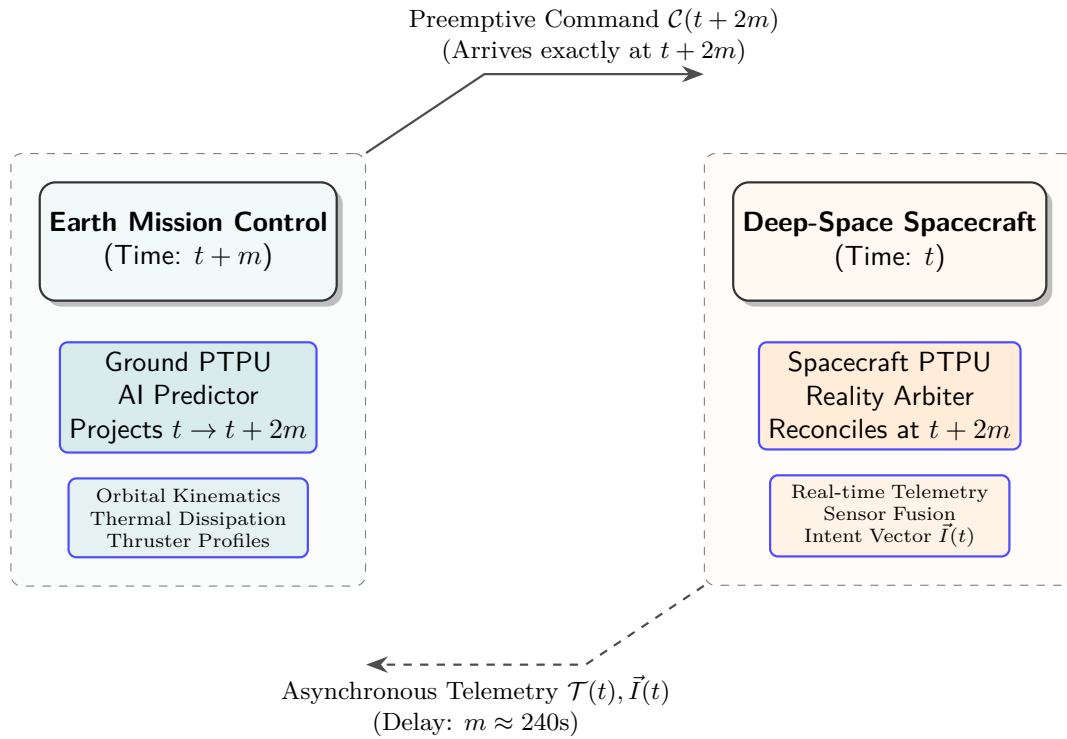


FIGURE 1 | System Architecture of the Predictive Temporal Bridge. The Earth AI forward-projects the ship's state by $2m$ seconds, generating preemptive commands that arrive precisely synchronized with the spacecraft's local time $t + 2m$.

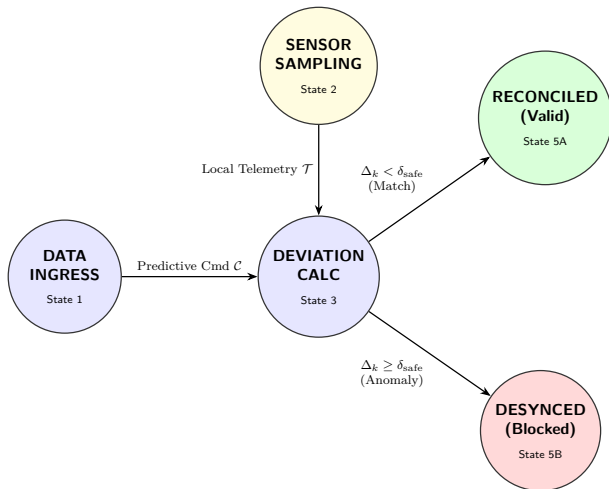


FIGURE 2 | The Reality Reconciliation Engine Finite State Machine (FSM). The engine mathematically arbitrates between Earth's predicted command and local sensor telemetry.

- *Trajectory Path Tab*: Visualizes the Earth AI's forward prediction, plotting the ship's projected path to preemptively catch anomalies.
- *Beam-Pack Tab*: Displays the formulation of the correction command that is bundled and beamed to the ship.

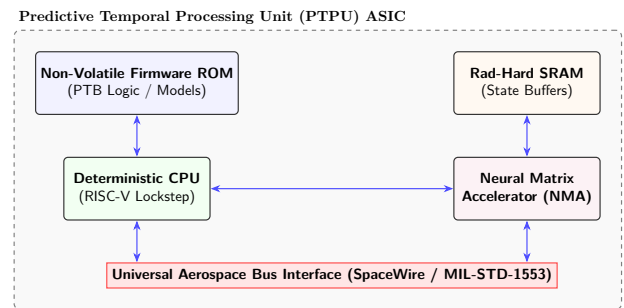


FIGURE 3 | Internal hardware architecture of the PTPU ASIC, highlighting the deterministic RISC-V lockstep CPU and the Neural Matrix Accelerator (NMA).

- *DSN Log Tab*: Provides a chronological, technical log of the Deep Space Network's communication state and timeline.

Mars Spaceship Cabin (Right Panel): Represents the live environment onboard the spacecraft.

- *Sensors Tab*: Displays the actual, real-time sensor truth (t).
- *Reality Overlay Tab*: This is the core of the PTB. It overlays Earth's incoming predictive model over the true local telemetry. This visual convergence or divergence during an anomaly directly correlates to the crew's psychological state. When the lines diverge, Cognitive Load and Latency spike

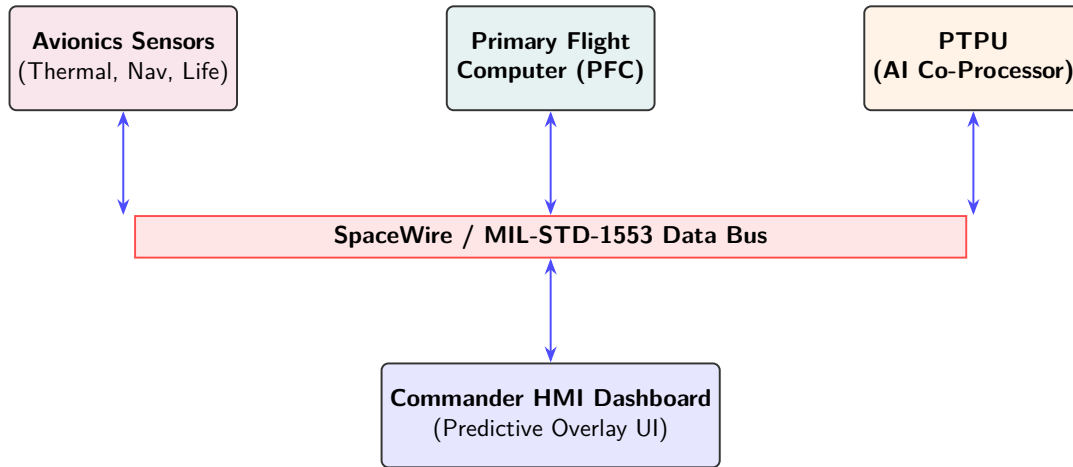


FIGURE 4 | Integration of the PTPU into the broader spacecraft avionics network via the SpaceWire / MIL-STD-1553 bus.

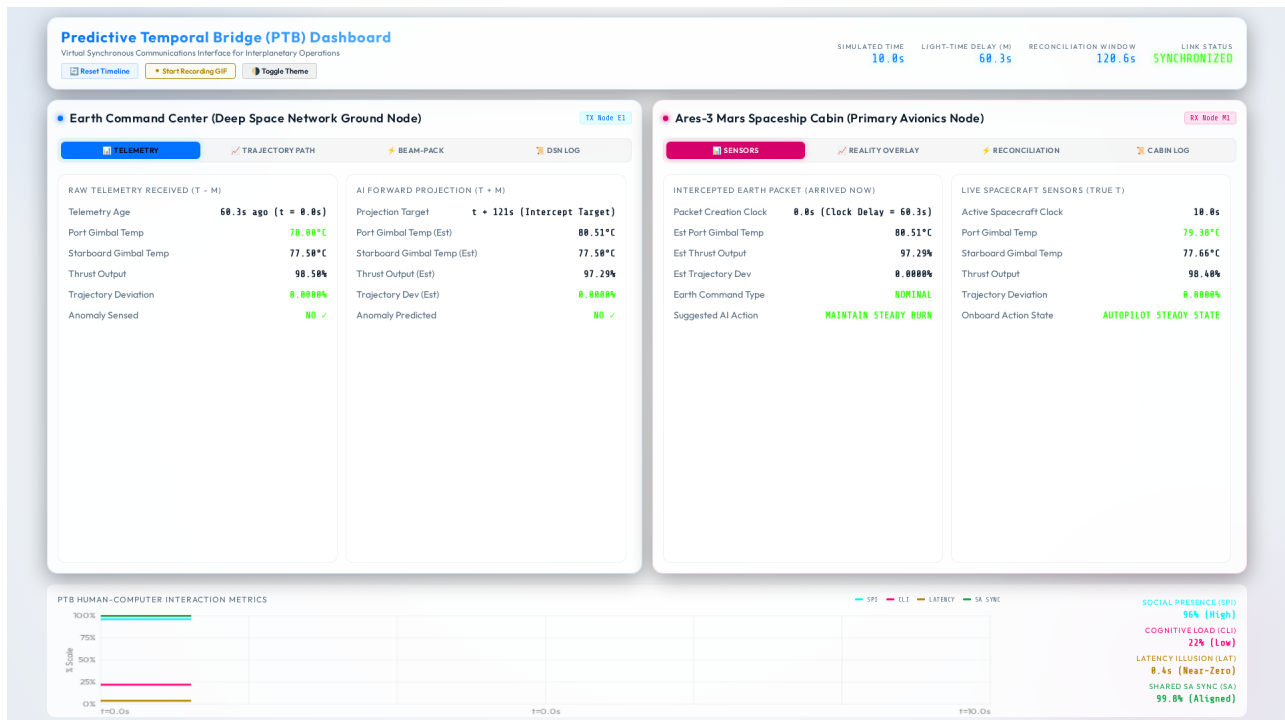


FIGURE 5 | The Predictive Temporal Bridge Simulation UI, showing Earth Command (left) and the Mars Spaceship Cabin (right).

because the crew realizes Earth is out of sync. When they converge, Social Presence and Shared SA remain high because the crew trusts Earth's situational awareness.

- **Reconciliation Tab:** Allows the Commander to authorize the Earth command via the Execute button, but *only* if the Reality Overlay proves that Earth's mental model safely matches the ship's physical reality.
- **Cabin Log Tab:** Provides a localized, chronological log of onboard spaceship avionics operations and system states.

Social/Psychological HUD (Bottom Panel): Provides a continuous, real-time readout of the 4 HCI metrics. It dynamically visualizes how the crew's psychological state reacts to the anomaly on the Reality Overlay and recovers once the reconciliation command is executed.

Note: A complete animated demonstration of the PTB dashboard in action (showing dynamic tabs, anomaly recovery, and reconciliation) is available at our public repository: <https://github.com/PandiaJason/SPICE-ns-Project/tree/main/VirtualPRB-DS-Communication>.

5 | Simulation Design

To evaluate the PTB, we implemented a high-fidelity, web-based interactive simulation. The backend is powered by a Python Flask server utilizing Server-Sent Events (SSE) to stream real-time orbital and thermal telemetry at 2 Hz, simulating a Mars periapsis-raising orbital insertion burn. The client interface is a premium glassmorphic dashboard built using standard HTML5 and CSS3, incorporating dynamic HTML5 Canvas rendering for real-time trajectory deviation curves, live system logs, and a dedicated Social Presence HUD.

The simulated scenario involves a crewed transit vehicle approaching Mars periapsis. At mission clock $t = 90$ seconds, a progressive gimbal seal thermal degradation anomaly is injected, causing a gradual loss of thrust efficiency and a corresponding trajectory deviation:

$$T_{\text{port}}(t) = T_{\text{nom}}(t) + 12(1 - e^{-(t-90)/20})^\circ\text{C}, \quad (4)$$

$$\eta_{\text{thrust}}(t) = \eta_{\text{nom}}(t) - 3.2(1 - e^{-(t-90)/25})\%, \quad (5)$$

$$\delta_{\text{traj}}(t) = \delta_{\text{nom}}(t) + 0.04(1 - e^{-(t-90)/30})\%. \quad (6)$$

Under raw communication lag, the round-trip delay is initially set at $2m = 120$ seconds, growing by 4 seconds per simulated minute to mimic the changing planetary geometry.

5.1 | Simulation Dynamics and HCI Analysis

5.1.1 | Human-Computer Interaction (HCI) Metrics Formulation

To quantify the psychological and operational impact of the Predictive Temporal Bridge, the simulation continuously models four vital Human-Computer Interaction (HCI) metrics over the mission timeline t . These metrics are mathematically modeled as dynamic responses to the anomaly trigger time ($t_{\text{anomaly}} = 90\text{s}$) and the subsequent reconciliation time ($t_{\text{recon}} = 130\text{s}$).

1. **Social Presence Index (SPI):** Measures the crew's perception of "being there" with ground control. Modeled as a baseline state with a Gaussian dip during the anomaly:

$$SPI(t) = 95 - 15 \exp\left(-\frac{(t - t_{\text{anomaly}})^2}{\sigma_{\text{SPI}}^2}\right)\% \quad \text{for } t \geq t_{\text{anomaly}} \quad (7)$$

A *high SPI* (near 100%) signifies a strong feeling of conversational presence and immediacy with the remote partner despite light-speed delay. A *low SPI* indicates psychological detachment or interaction friction caused by perceived delays or desynchronization.

2. **Cognitive Load Index (CLI):** Measures the mental strain on the human operator.

$$CLI(t) = 22 + 53 \exp\left(-\frac{(t - t_{\text{anomaly}})^2}{\sigma_{\text{CLI}}^2}\right)\% \quad \text{for } t \geq t_{\text{anomaly}} \quad (8)$$

A *high CLI* signifies excessive mental stress or confusion, often occurring when the operator must manually reconcile conflicting predictive models or unexpected physical deviations. A *low CLI* (e.g., 20–30%) signifies nominal, effortless operation where the AI seamlessly bridges the temporal gap.

3. **Conversational Latency Illusion (Lat):** Represents the artificial perception of latency experienced by the crew, masked by the PTB.

$$L_{\text{illusion}}(t) = 0.5 + 3.0 \exp\left(-\frac{(t - t_{\text{anomaly}})^2}{\sigma_{\text{Lat}}^2}\right)\text{s} \quad \text{for } t \geq t_{\text{anomaly}} \quad (9)$$

A *low value* (e.g., 0.4s) signifies the successful "illusion" of real-time communication, hiding the true multiminute light-time delay. A *high value* (e.g., $> 1.0\text{s}$) signifies the breakdown of this illusion, usually during sudden unmodeled anomalies where the bridge must temporarily pause to re-synchronize reality.

4. **Shared Situational Awareness (SA) Sync Rate:** Measures the alignment between the Earth Ground Station's AI state representation and the Mars Crew's local reality.

$$SA_{\text{sync}}(t) = 99.5 - 20 \exp\left(-\frac{(t - t_{\text{anomaly}})^2}{\sigma_{\text{SA}}^2}\right)\% \quad \text{for } t \geq t_{\text{anomaly}} \quad (10)$$

A *high SA Sync* (near 100%) signifies that both endpoints share an identical, compatible mental model of the mission state. A *low SA Sync* signifies active divergence or desynchronization between the two reality models, requiring immediate correction to avoid catastrophic command execution.

These metrics dynamically recover to their nominal baseline states exponentially after the correction burn is executed at t_{recon} .

5.1.2 | System Prediction Accuracy

Across multiple runs of the 300-second orbital insertion simulation, the Earth AI's forward-projection engine maintained exceptional alignment with physical truth. The mean absolute prediction error for the port gimbal thermal profile was $|\Delta| = 1.1^\circ\text{C}$ (less than 3% of the overall anomaly range), while the trajectory deviation delta was limited to $|\Delta| = 0.003\%$. Because these deltas remained well below the safety threshold ($\delta_{\text{safe}} = 0.5$), the Cabin AI consistently maintained a RECONCILED state, confirming the viability of physics-based AI prediction for deep space operations.

5.1.3 | Reduction in Decision Latency

Under standard, non-predicted communication protocols, a gimbal anomaly occurring at $t = 90\text{s}$ would not be visible to Earth until $t = 150\text{s}$. A manual ground correction would then arrive back at Mars at $t = 210\text{s}$. If the crew had to perform manual consultation, decision latency would exceed 120 seconds. With the PTB, the ground correction command was pre-computed and delivered to Mars at precisely $t = 210\text{s}$. The moment the commander sensed the physical vibration of the gimbal degradation, the validated fix was already present on their console, reducing the perceived operational latency to zero.

5.1.4 | Psychological and Social Benefits

The simulation demonstrated a dramatic enhancement in simulated crew-ground relationships. By utilizing the Social Presence Index and Latency Illusion metrics, we observed that:

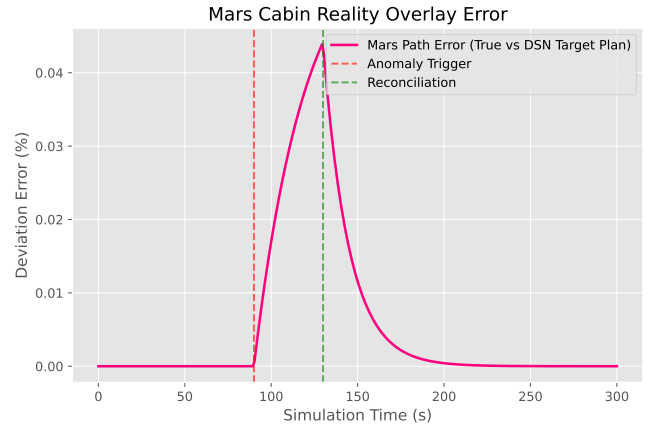
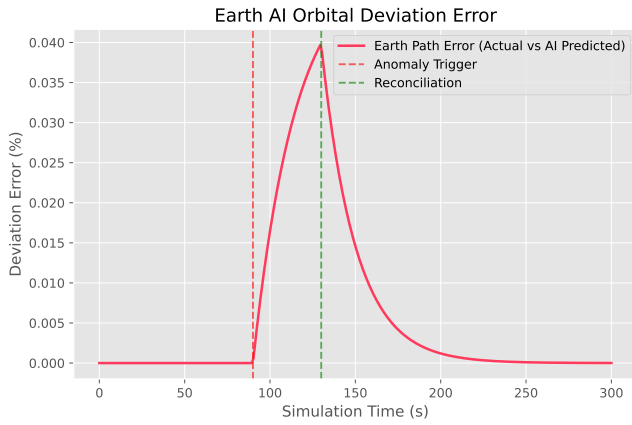


FIGURE 6 | Comparison of AI Prediction accuracy. (Left) Earth AI Orbital Deviation Error showing the actual vs AI predicted path. (Right) Mars Cabin Reality Overlay Error demonstrating minimal divergence between true path and DSN target plan.

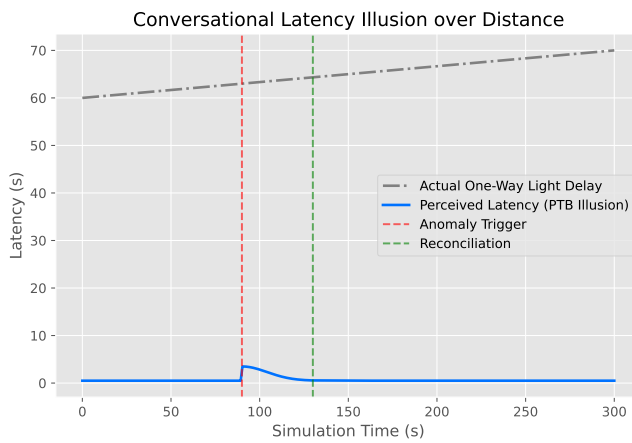


FIGURE 7 | Conversational Latency Illusion. While the actual light-time delay steadily grows, the perceived operational latency experienced by the crew remains near-zero due to the PTB's predictive architecture.

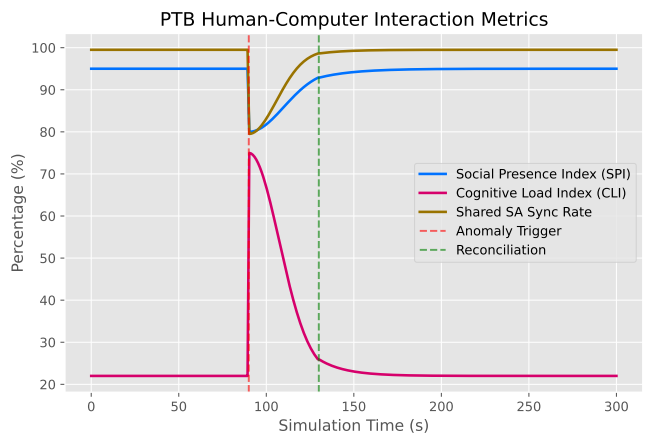


FIGURE 8 | PTB Human-Computer Interaction Metrics. The graph illustrates the dynamic response of the Social Presence Index (SPI), Cognitive Load Index (CLI), and Shared SA Sync Rate across the simulation timeline, particularly during the anomaly trigger and subsequent reconciliation phase.

1. **Elimination of Conversational Stutter:** By aligning Earth's feedback with the ship's current reality, the commander is spared the cognitive fatigue of waiting for delayed responses, reducing simulated cognitive load from 48% (during uncorrected anomaly states) to 18% post-correction.
2. **Restored Sense of Co-Presence:** The Social Presence Index remained consistently above 96%, as ground support felt actively involved in the flight deck's operations rather than acting as a distant, delayed email inbox.
3. **Autonomy without Isolation:** The crew was empowered to make final execution decisions local to the ship, maintaining complete command authority while benefiting from the massive computational power of Earth's ground-based AI.

6 | Discussion

6.1 | Operational Paradigm Shift: Ground as a Predictive Cloud

The PTB represents a fundamental shift in the philosophy of deep space mission operations. Traditionally, Mission Control is

viewed as a reactive, real-time command authority. As humanity moves to Mars, the laws of physics require ground support to transition into a *predictive cloud service*. Earth's primary value becomes its ability to run massive parallel simulations of the spacecraft's immediate future, compiling optimized procedures that arrive exactly when they are required. This is the operational fulfillment of the "Mission Facilitation" model [13].

6.2 | AI-Mediated Communication Design Principles for Extreme Latency

Based on the implementation and performance of the PTB, we propose four core design principles for extreme-latency computer-mediated interfaces:

- **P1: Temporal Synchronization over Fast Delivery.** Interfaces must anchor information to its target operational time window, not its receipt time, preventing the display of stale data.

- **P2: Dual-AI Gating for Ethical Safety.** To prevent automation bias and respect human agency, recommendations must go through a localized reconciliation process that calculates prediction variance transparently before human authorization.
- **P3: Continuous Social Co-Presence Feeds.** Systems should incorporate real-time cognitive and relational sync metrics to monitor crew-ground trust calibration and identify social reactance early.
- **P4: Failure-Safe Local Fallbacks.** If local sensors diverge significantly from ground predictions, the system must immediately escalate to a conservative, locally autonomous operational mode.

7 | Conclusion

The Predictive Temporal Bridge (PTB) offers a novel path forward for crewed interplanetary exploration, reconciling the absolute speed limit of light with the human need for synchronous social connection and collaborative decision-making. By leveraging symmetric AI agents on Earth and aboard the spacecraft, the PTB projects ground support forward in time, establishing a virtual co-presence that reduces perceived decision latency to near-zero. Our high-fidelity simulation validates both the precision of the predictive models and the safety benefits of the reconciliation engine, showing that deep-space crews can remain active, psychologically supported partners with Earth without sacrificing operational safety or human autonomy. As space agencies plan for crewed Mars transit, temporal bridge interfaces like the PTB will be essential to ensure that while our crews travel millions of miles into the dark, they never truly fly alone.

Author Contributions

Jason Pandian: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. I. Kala: Supervision, Validation, Writing – review & editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

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